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Ground Water Quality Analysis with Real-Time Data Using Machine Learning

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Abstract – For the entire world Ground water is considered as the predominant source of water used for various purposes. As far as India is concerned 65% of its needs are satisfied by ground water. But it is polluted with improper disposal of solid and liquid substances which leads to the presence of heavy metals and noxious elements. Several technologies are used to envisage water quality. With respect to change in climate factor, change in temperature, heavy contamination the accuracy would not attain high. Hence it is decided to adopt Machine learning technique in order to predict standard of water quality. Different Algorithms are used to train the model and the accuracy of all algorithms is then compared to check which algorithm has the highest accuracy to predict the quality of the water.

Keywords: Ground Water, Machine Learning, ANN, Real time data, Support Vector Machine (SVM)

I-INTRODUCTION

The article [1] compares the water quality from different water stations. Here, fifty sampling stations that were identified for gathering water samples of groundwater quality. The water sampling in this case was useful for determining the quality of the groundwater, but the two other stations were not suitable for human consumption[2]–[5]. Several physio-chemical characteristics were used to describe the quality of ground water. Here, the town of Visakhapatnam began to grow in significance for the East India Company. Visakhapatnam Municipal Corporation, this one was outstanding. The idea behind the water quality index is to show the progression of water quality. It serves as a representation of the contribution element. Ground water is very important for the sustainable development of any country. [6], [7] classify the ground water sample based on the characteristics of the water sample. Based on the analysis, it has been found that around 47% of the samples, according to the ground water quality index (GWQI), were considered to be of good quality for drinking. The majority of the samples fall into the low level of pollution category according to the heavy metal pollution index (HPI), degree of contamination (Cd), and heavy metal evaluation index (HEI). [8]–[10] briefly describes how the groundwater is an important parameter for the development of any country. There are many ways to determine the ground water quality. GIS [11] is a statistical technique which can be used for the evaluation of Ground-water. Living creatures in water are desirable, which is reflected in a sometimes-significant price premium for real estate. It smelled like surfing, swimming, and fishing in this water quality of surface values[12].

Become healthier, happier, less stressed, and more peaceful when we are in the water. Wisconsin was the site of early US investigations on the connection between property values and water quality. 13 out of the 12 studies from the middle of the US's history that are reviewed were carried out there. The Netherlands' central region was the subject of international studies looking at the impact of water quality on real estate prices in 2005. The studies [13] that were examined all consistently showed a connection between property value and surface water quality. The review's conclusions have significant implications for environmental policy, including maintaining greater quality value.

II-REVIEW OF LITERATURE

Agriculture is the back-bone of our country and groundwater is the primary source for Agriculture. Because of the over dependence of Fluorine in water, it drastically affects the quality of water[14]–[18]. Many suggest that Fluorine is good for health but high consumption of fluorine can lead to several health issues like Seizure. Most of the pollution in Ground-Water causes because of Man-Made Activities such as Mining Which produce Trace Metals (TM) which creates a major global concern because of its high toxicity to surrounding environments and can also have serious health effects like cancer to humans via

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multiple pathways [19]. Because of long residence of ground water, the trace of micro-organisms can persist for decades[20]–[23]. Similar to Micro-Organism, Micro-Plastic is also detected in large amounts in water. Micro-Plastic are the new major environmental concern that consists of plastic particles of size less than 5mm which pollutes the water via waste-water and large litters of plastic of small size[24]. Another source of ground-water contamination is pesticides. Pesticides are the chemicals that are used to kill pests like insects, fungi etc., from plants.

There are thousands of pesticides that are available in the market and it contains harmful chemicals that not only can kill pests but also humans. The residue of pesticides remains in the soil after its use and is then mixed with soil through various pathways along with other harmful chemicals present in the soil. This not only makes the water toxic but can even kill humans and animals when consumed[15], [25], [26]. Gases like CO_2 when leaked affects the pH and turbidity of the water thereby changing the entire nature of the water [27] .

Because of all the chemicals present in groundwater from various sources it then combines together to form uranium. The Uranium Threat comes in it chemical form rather than its radioactive form [28], [29]. Also, previous researches and this research also prove that in some parts of India. The Uranium Concentration is much higher than the limit provided by World Health Organisation (WHO). Exposure to Uranium in short-period will not have any negative impact however when exposed on a long-period can affect bones and Kidneys[28].

III-METHODOLOGY

A. Machine Learning Solution

Few years back, there are other ways to predict the quality of water. But due to climate change and various other factors, the accuracy will not be high. But with the help of machine learning, this is possible. Machine learning be allied to Artificial Intelligence (AI) which is used to predict several things in several industries. There are several Algorithms that are used in Machine Learning and each algorithm has its own unique use and advantages[30].

Agriculture sector is one the biggest and most important sector in India. As a matter of fact, India is one of the biggest agricultural exporters in the world. Several climatic factors, it is impossible to predict the quality of crops in every yield. In order to prevent this, Machine Learning algorithm is developed to predict the requirement of crops during every cultivation[31]. It considers several parameters like climatic, environmental and cultural needs to predict the requirement for crops. It also predicts the possible disease that a crop can get. As a further advancement in this domain, this article [32] developed an algorithm that also predicts the rainfall rate and soil characteristics using a special Machine Learning Algorithm called Artificial Neural Network (ANN). Its also to be noted that using ANN, the algorithm is able to predict the parameters more accurately compared to other algorithms. Quality of crops is not the only concern in the Agricultural. Soil Water Erosion (SWE), similar to crops quality, is also a critical concern in the Agricultural Sector.

Soil Water Erosion is defined as the displacement of upper-most layer of soil and water through various factors such as wind and water. In agriculture, soil erosion causes uneven distribution of soil which might affect the nutrients that the crop might get. Apart from agriculture, when mixed with water, Soil Erosion can have huge impacts like increasing the water turbidity level which in-turn will decrease the water quality. In this article [33] several algorithms are used to predict the accuracy, where it has been found that ANN and ANFIS have higher prediction accuracy when compared to other algorithm.

IV-RESULT AND ANALYSIS

A. Data Information

Water sample is collected at Somangalam area but this model is trained with the dataset of water sample collected at study area. It contains approximately 12000 datasets containing parameters like Turbidity, Temperature, pH, TDS, DO and depth of the water. The dataset has been collected using IoT Kit and has been collected during the post monsoon period.

B. Data Cleaning

Data Cleaning is the first and important step in creating a model that uses Machine Learning Technology. Data Cleaning is the process of removing missing or misleading data which might affect the efficiency of the model. Data Cleaning can be done in two ways[34]

The first method is to remove the missing data which gets rid of the missing data but also reduces the quantity of data which in-turn affects the accuracy of the model[35] The second method is to replace the missing data with the average value of the data. In this way, not only the quantity of the dataset will not be affected, the dataset will also be unbiased.

After examining the dataset, it has been found that around 2500 dataset were found missing. These missing datasets are replaced with the average value of the data. In this way, both the quantity of the dataset and the accuracy of the model will not be affected.

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```
Turbidity (0-3000 NTU)     2544
TDS (0-1000 PPM)           2545
Temperature (ºC)           2544
pH (0-14)                  2544
DO (0-20 mg/L)             2544
Depth of water(ft)         6919
dtype: int64
```

FIG 1 WATER QUALITY DATASET

C. *Exploratory Data Analysis*

Exploratory Data Analysis is similar to Data Cleaning process which is another important step in creating a efficient model. Exploratory Data Analysis is the process of further examining the dataset using advanced tools to gain more insight on the data[36]

TABLE 1 EXPLORATORY DATA ANALYSIS

	Turbidity (0-3000 NTU)	TDS (0-1000 PPM)	Temperature (ºC)	pH (0-14)	DO (0-20 mg/L)	Depth of water(ft)
count	11449.000000	11449.000000	11449.000000	11449.000000	11449.000000	11449.000000
mean	64.571280	678.188914	29.047455	9.878258	-4.075398	9.505690
std	236.808233	272.829732	1.293510	1.787198	17.337285	1.498804
min	0.000000	0.000000	23.169000	5.800000	-32.750000	7.000000
25%	0.000000	626.810000	28.176000	9.380000	-18.490000	9.505690
50%	0.000000	678.188914	29.047455	9.878258	-4.075398	9.505690
75%	64.571280	859.940000	29.169000	11.870000	11.380000	9.505690
max	3000.000000	1347.980000	40.169000	13.650000	32.750000	15.000000

D. *Heat Mapping*

Heat Mapping is one of the most advanced tools which is used to compare different parameter with each other. Heat mapping of the dataset has been attached below:

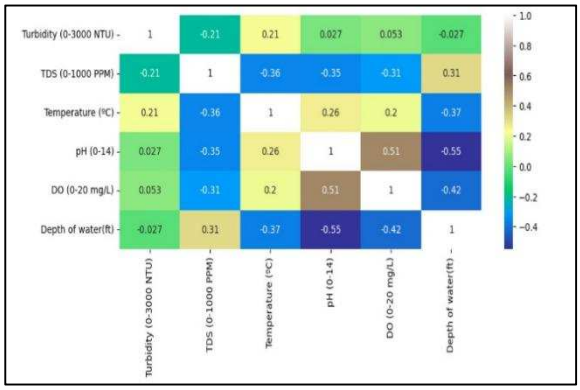


FIG 2 – HEAT MAPPING

From the above heatmap, it is found that almost 50% of the time, change in the DO, also changes the pH of the water and vice-versa.

E. *Train-test model*

For the purpose of training and testing the model, Eighty percent of the dataset (Approx. 9000 datasets) is used to train the model and the outstanding twenty percent of the dataset is practiced to test the model.

F. *Algorithms Used*

Algorithms applied in the Machine Learning is well-defined as the method used by the Artificial Intelligent system to convey out its task—usually, envisaging output from given input data. Classification and also regression are the two basic techniques used by machine learning systems. In this project, the algorithms used in this project are mentioned below

1. Decision Tree Classified
2. KNN Algorithm
3. ANN Algorithm
4. SVM Algorithm

G. *Decision Tree Classifier*

The decision tree classifier is a supervised learning method that can be used to solve classification and regression problems, but is usually preferred for this purpose. It is a tree-structured classifier where the internal nodes represent features of the dataset, the branches represent decision-making processes, and each leaf node represents an outcome. The decision tree classifier has several parameters that are adopted by default, which can also be tuned to improve accuracy.

In Decision Tree Classifier, two different tests are done. First test is done using the default criteria and the second test is done using the best criteria. In both the test, Eighty percent of the dataset is used to train the model and the remaining twenty percent of the dataset is utilised to assess the model.

On the first test, with the default criteria, an accuracy score of 97.3% is obtained and on the second test, with the best criteria, an accuracy score of 96% is obtained.

H. *KNN Algorithm*

The K-Nearest Neighbors algorithm which is also identified as KNN or k-NN is a administered learning classifier that belongs to proximity to make groupings or predictions about alliances of single data points. The amount of dataset is divided into Eighty-Twenty where Eighty percent of the data-set is utilised to train the KNN algorithm and twenty percent of the dataset is taken to test the algorithm. Subsequently a precision score of 95.5% is obtained

I. *ANN Algorithm*

ANN also known as Artificial Neural Network is a special algorithm that can be used in machine learning project. ANN is often referred as Human brain in computer form. Almost most of the time, the model which is trained using ANN is successful. In order to train the ANN, eighty percent of the dataset is used for training the ANN model and the remaining twenty percent of the dataset is used for testing and crosschecked to determine the accuracy. After testing, an accuracy score of 91% is obtained

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J. SVM

SVM technique is a powerful machine learning technique which can accurately categories or predict data points while necessitating little computational effort. Eighty percent of the dataset are utilised to train the SVM and tested the algorithm with residual twenty percent dataset. Hence the accuracy score of SVM technique is 77.6%

K. AdaBoost Classifier

The meta-estimator, which is also called the AdaBoost classifier, integrates a classifier to a preliminary data set. Later applies a supplementary copy of the classifier to the similar data set, nevertheless reweights the mis-classified instances so that later classifiers focus more on difficult cases. AdaBoost achieves an accuracy value of 75.2%.

L. Accuracy Comparison

The accuracy score obtained from the algorithm mentioned above is converted into a bar plot and been attached below for better understanding

TABLE 2 ACCURACY COMPARISON

	Model	Accuracy Score
0	Decision Tree	0.962882
1	ANN	0.909607
2	KNN	0.955459
3	SVM	0.775983
4	AdaBoost	0.752838
5	r2_score	0.692113
6	Mean Absolute Error	0.656687
7	Mean Squared Error	0.796475

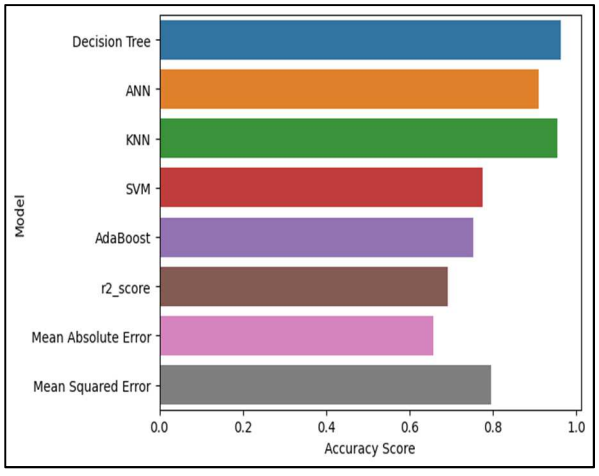


Fig.3 – Accuracy Score Comparison

V. SUMMARY AND CONCLUSION

Machine learning technology is used to determine the quality of water. Numerous Algorithms like ANN, SVM, KNN are tried to train the model. From Somangalam tank of Kancheepuram district water sample was collected and tested for various parameters. The model has been trained on the water parameters splitted into 80-20 ratio. That is, Eight percent of the datasets are used to train the model and the remaining twenty percent has been used to test the model. It has been found that the ANN algorithm is predicting correctly with better accurateness when compared to other algorithms based on the accuracy score received and after comparing each accuracy score,

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